Prince Sultan University
College of Computer and Info Sciences / Department of Information Systems
Semester – AY(2026-2027)

COURSE SYLLABUS

1. Course number and name: CYS401: Fundamentals of Cybersecurity
2. Credits and contact hours: 3 (3,0,1)
3. Instructor's name: Dr. Basmh
    a. Scheduled Office Hours: TBD
    b. Office Location: W311
    c. Email: balkanjr@psu.edu.sa

4. Text book, title, author, and year
    a. Primary Text: Cybersecurity Essentials Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short, (2018)
    b. Other References:
        - Bhusan, M., Rathore, R. S., & Jamshed, A. (2018). Fundamental of Cyber Security: Principles, Theory and Practices. BPB Publications.
        - CISSP: Certified Information Systems Security Professional Official Study Guide, 8th Edition, by Mike Chapple, James Stewart and Darril Gibson, ISBN-13: 978-1119475934, SYBEX, Wiley Brand.
    c. Electronic Materials: None
    d. Learning Management System: Moodle available at https://lms.psu.edu.sa

5. Specific course information
    a. Brief description of the content of the course (catalog description):
    This course serves as an introductory course to the Cybersecurity track whereby the basic concepts including cybersecurity in the business context, threats and attacks in different systems, cybersecurity authentication and authorization techniques, cybersecurity models and approaches, and the latest trends in cybersecurity. Besides, the course is designed to provide an overview and understanding of established cybersecurity strategy and offers students the opportunity to engage in strategic decision making in the context of cybersecurity.
    b. Prerequisites or corequisites: None
    c. Indicate whether a required, elective, or selected elective (as per Table 5-1) course in the program: Required

6. Specific goals for the course
    a. Specific outcomes of instruction. The student will be able to:
    CLO 1: Discuss the fundamental issues and techniques of cyber security.
    CLO 2: Analyze the security models and approaches for specific security needs.
    CLO 3: Distinguish the security principles and their legal, ethical and professional aspects.
    CLO 4: Apply appropriate cybersecurity countermeasures to prevent, detect, react, and recover from simple attacks.

b. Explicitly indicate which of the student outcomes listed in Criterion 3 or any other outcomes are addressed by the course.

Course LOs #	**Student Outcomes**		
	Computer Science	**Software Engineering**	**Information Systems**
1	1	1	1
2	2	2	2
3	4	4	4
4	6	6	6

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Brief list of topics to be covered

Week No	Ch No.	Topics	CLO	Assessment
1	**1**	**Introduction to Cybersecurity** ○ Cybersecurity concepts ○ Cybersecurity objectives and Goal. ○ Cybersecurity Triad and System components ○ Critical Information characteristics in Cybersecurity	**CLO 1**	Lab & Class Activities
2	**2**	**Security Foundations and Principles** ○ Cybersecurity foundations and elements ○ Threat, Attack, Vulnerability, and countermeasures in Cybersecurity ○ Attacks Types in Cybersecurity and their countermeasure ○ Threats Modelling and STRIDE ○ Protection Mechanisms and Countermeasures ○ Cybersecurity Governance and Planning in Business	**CLO 2, CLO 4**	Lab & Class Activities
3-4	**3**	**Data Security and Information Asset Protection** ○ Information Asset and Information Asset Domain ○ Due Care vs Due Diligence ○ Information Classification and Classification Procedure ○ Governmental Classification vs. Commercial Classification ○ Data Labeling and Protection Mechanisms ○ Data Sanitization Techniques ○ Data ownership roles and responsibilities. ○ Data Protection Legislation and Regulations (GDPR, PDPA, etc)	**CLO 2, CLO 3**	Lab & Class Activities, Quiz 1
5-6	**4**	**Cryptography Fundamentals.** ○ Basic Concepts of Cryptography ○ Cryptography Components and process ○ Symmetric vs. Asymmetric Cryptography ○ Digital Signature and Data Integrity ○ Public Key Infrastructure and Digital Certificate	**CLO 1, CLO 4**	Lab & Class Activities
7	**5**	**Principles of Security Design** ○ Security Architecture principles ○ Common Architecture Flaws and Security Issues ○ Protection Mechanism for Ensuring Confidentiality, Integrity, and Availability: Bounds, Process Isolation, and Confinement	**CLO 1, CLO 2**	Lab & Class Activities, Major Exam
8	**6**	**Computer Architecture Security** ○ Hardware Components & Protection Rings ○ Memory Protection Mechanisms ○ Firmware and Operating System Hardening ○ Input/Output Security and TEMPEST ○ Server- client Systems Security: Server-side Attacks and Client-side Attacks	**CLO 1, CLO 2**	Lab & Class Activities
9	**7**	**Systems Architecture Security** ○ Database Systems Security ○ Cloud Computing and Virtualization with security Aspects ○ Internet of Things security ○ Web-Based Systems security and OWASP Program ○ Mobile Devices Security	**CLO 1, CLO 4**	Lab & Class Activities, Quiz 2
10-11	**8**	**ICS/SCADA System Security for CPS** ○ ICS/SCADA Systems Security ○ Vulnerability ○ Threats, challenges & countermeasures ○ Security requirements ○ Policies ○ Governance and Compliance ○ ICS/SCADA systems security in smart grid	**CLO 1, CLO 4**	Lab & Class Activities
12-13	**9**	**Managing Identity and Authentication** ○ Access Control Models ○ Access Control Attacks ○ Controlling Access to Assets ○ Identification and Authentication ○ Implementing Identity Management	**CLO 1, CLO 4**	Lab & Class Activities, Final Exam

		○ Biometric standards ○ Biometric applications ○ Authentication vs. Identification		
14		**Project Presentations**	**CLO1,2,3,4**	Project
15		**Revisions**		

8. Weight of Assessments

No.	Assessment Type	Weightage	Date	Time
1	Quiz1	5%		12-1
2	Quiz 2	5%		12-1
3	Major	20%		12-1
4	Labs & Activities	10%	Throughout semester	Class time
5	Attendance	5%	Throughout semester	Class time
6	Project	15%	Week 13 (submission)	Week 14 (presentations)
7	Final Exam	40%		12.30 pm